

Questions to Measure Subjective States

This chapter addresses the measurement of people's subjective states: their knowledge and perceptions, their feelings, and their judgments. As noted previously, a distinctive feature of the measurement of subjective states is that there are, in fact, no right or wrong answers to questions. "Rightness" implies the possibility of an objective standard against which to evaluate answers. Although the consistency of answers with other information can be assessed, there is no direct way to know about people's subjective states independent of what they tell us.

That statement does not mean that there are not standards for questions designed to measure subjective states. The standards are basically the same as for questions about factual things: questions should be understood consistently by all respondents so they are all answering the same question, and the response task, the way respondents are asked to answer the questions, should be one that respondents can do consistently and that provides meaningful information about what they have to say.

By far, the largest number of survey questions ask about respondent's perceptions or feelings about themselves or others. The basic task of most questions in this category is to place answers on a single, well-defined continuum. For descriptive questions, a dimension is defined such as hot to cold, slow to fast, or frequent to infrequent. Respondents are asked to place themselves, or what they are rating, on that dimension. If the question is about judgments or feelings, the rating dimension will be some variation of positive to negative. The majority of this chapter will be devoted to the design of such questions.

Questions designed to measure respondent agreement with or support of ideas raise some different issues, as do questions designed to measure knowledge. Separate sections describing those special issues are included at the end of the chapter.

DESCRIBING AND EVALUATING PEOPLE, PLACES, AND THINGS

The following are some examples of question objectives. They are not intended to be well worded; they are intended to define the general informational goals that questions might be designed to achieve.

Example 3.1a: How friendly is your doctor?

Example 3.1b: Do you consider the president to be liberal, conservative, or somewhere in between?

Example 3.1c: How does the level of crime in this neighborhood compare with crime rates in other neighborhoods in the city?

Some questions in this category might seem to belong in Chapter 2, which concerned the reporting of objective events. For example, the answer to a question about crime rates could be thought of either as an objective measure of crime rates or as a measure of people's perceptions. Although people may agree on the friendliness of a particular physician, the liberalness of a president, or the level of crime, these properties are in the eyes of the beholders. If a respondent says a doctor is not friendly, the answer constitutes its own truth, even if others have a different opinion.

Questions about self perceptions are also covered here. Common examples include:

Example 3.2a: How frequently are you happy?

Example 3.2b: How interested are you in political events?

Example 3.2c: How likely are you to vote?

These are all questions that ask respondents to look at themselves, pull together information they have about themselves, and provide a summary of some sort. In addition to asking people to describe themselves, researchers also ask people to put those perceptions up against some standard, either an objective standard or a self-imposed standard.

Example 3.3: Would you say your current weight is too much, about right, or too little?

All of the objectives outlined above would be achieved by questions that have the same basic framework, which consists of three components.

1. What is to be rated
2. What dimension or continuum the rated object is to be placed on
3. The characteristics of the continuum that are offered to the respondent.

Defining What Is to Be Rated

Like all survey questions, when designing questions such as this, it is important that everyone be answering the same question.

Example 3.4: How would you rate your health—excellent, very good, good, fair, or poor?

This question probably has been asked in more health surveys than any other single question, and it has frequently proven to be a valid and useful measure of health status. Nonetheless, it may not be surprising to learn that the meaning of “health” is not consistent across all respondents. Some people key on the presence of health conditions, whereas others key on the extent to which they feel fit or the extent to which they lead healthy lives. Although these concepts may be interrelated, it is undesirable to have people answering different questions. To the extent that the meaning of health could be standardized across respondents, the question would have less variance attributable to interpretation of the question and, consequently, more true variance associated with the perceptions of the respondents answering the question.

Example 3.5: In general, do you think government officials care about your interests a lot, some, only a little, or not at all?

“Government officials” are a very heterogeneous lot, and which government officials a respondent has in mind may affect how the question is answered. For example, people consistently rate local governments as more responsive than state and federal governments. Elected officials may not be rated the same as those in the executive branches of government. To the extent that people’s answers vary based on the

way they interpret the question, a new source of error is introduced into answers, and the answers will be less good measures of what the researchers are trying to measure.

Example 3.6: Do you consider crime to be a big problem, some problem, or no problem at all?

Crime is also heterogeneous. Can people lump white collar crime, drug dealing, and armed robbery into a single integrated whole? It would not be surprising if respondents keyed on different aspects of crime. Moreover, this particular question does not specify a locus for the problem: the neighborhood, the city, the local region, or the nation. Which perspective people take will affect the answers. People generally rate the crime problems in their own neighborhoods as less severe than average. To the extent that what is being rated can be specified more clearly, so that respondents do not vary in their interpretation of what they are rating, measurement will be better.

The Response Task

Researchers have designed numerous strategies for evoking answers from respondents. The most common task is some variation of putting the object of the answer on a continuum. In addition, respondents may be asked to:

1. answer in an agree-disagree form
2. rank order several objects
3. answer in narrative or open-ended form
4. use magnitude estimation techniques

Each of these alternative response tasks will be discussed, starting with a straightforward rating task.

Defining the Rating Dimension. Figure 3.1 defines a continuum from positive to negative. Such a continuum can be described to people in numerous ways, and there are numerous ways people can be asked to assign answers to a position on that continuum.

Example 3.7a: Overall, how would rate your health—excellent, very good, good, fair, or poor?

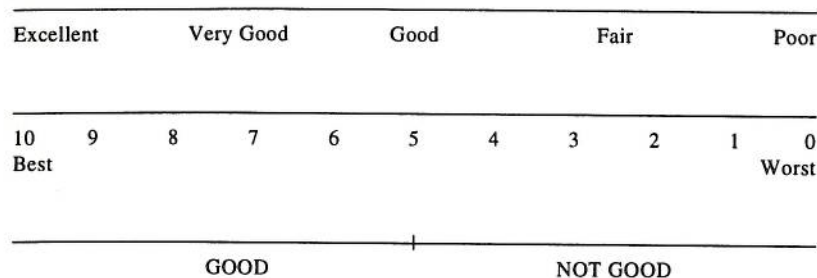


Figure 3.1. Some Examples of Evaluative Continua

Example 3.7b: Consider a scale from 0 to 10, where 10 represents the best your health can be, where 0 represents the worst your health can be, and the numbers in between represent health states in between. What number would you give your health today?

Example 3.7c: Overall, would you say you are in good health?

The above three questions are all using the same continuum. They are all asking the same question. They differ only in the ways in which the continuum is used.

In the first question, the health scale is divided into 5 categories, and each category is labeled with an adjective, from excellent to poor. The second question uses the same scale, but only the extremes of the scale are labeled; the respondents are allowed to use 11 different response categories, rather than 5; none of the middle categories is labeled with an adjective; and the respondents are free to use the numbers between 0 and 10 in any way they want. However, in both cases, there is a clear order among the response alternatives. "Very good" is better than "good" in the first question; eight is better than six in the second question.

The third question is a variation on the same theme. The continuum is divided into 2 categories, rather than 5 or 11. The positive category is labeled "good," and the negative category is labeled only implicitly as "not good."

These various differences among the questions have implications for resulting measurement. All of these questions define a single dimension, and they specify for all respondents a reasonably clear response task. Moreover, the fundamental assumption of ordinal measurement,

that people putting themselves in a higher category on average are giving a more positive response than those who put themselves on a lower category, is very likely to be met.

Whether a continuum is labeled with adjectives or with numbers, there obviously is potential for people to use the scales differently. The level of health that one person considers to be "good" may be considered to be only "fair" by another person. To the extent that people differ in the way they use these scales, there will be error in the measurement. Anything that affects responses, other than the way that people see the things that they are rating, reduces response validity.

A critical criterion for a response task is that it define a single dimension, and that the categories of responses from which respondents choose have a clearly ordered component.

Example 3.8: How would you say you feel about your lawn? Would you say:

Very satisfied
Somewhat satisfied
Satisfied
Not satisfied

Comment: There are many observers who would say that "somewhat satisfied" is a lower, more negative category than satisfied. If some respondents concur in that judgment and use "satisfied" as the second most positive category, the ordinal assumption about responses would be broken, and it would create serious unreliability in the measurement process. If adjectival labels are to be used, it is critical that their order be unambiguous.

Example 3.9: Which category best describes your physician?

Very competent and businesslike
Very competent and friendly
Fairly competent and friendly
Fairly competent and not friendly

Comment: There obviously are two dimensions in the response categories: competence and friendliness. This particular set of response categories did not even play out all of the possibilities on the two

dimensions in a systematic way. As a general rule, having respondents deal with two dimensions at a time is a mistake. Good survey design will ask respondents to place answers on a single, well-defined dimension.

Characteristics of Categories or Scales. When the goal is to have respondents place themselves or something else along a continuum, there are choices to be made about the characteristics of the scale or response task that is offered to respondents. Two key issues include:

1. How many categories to offer
2. Whether to use scales defined by numbers or by adjectives

In general, the goal of any rating task is to provide the researchers with as much information as possible about where respondents stand compared to others. Consider the continuum from positive to negative, such as portrayed in Figure 3.1, and the results of a question such as:

Example 3.10: In general, would you rate the job performance of the president as good or not so good?

Such a question divides respondents into two groups. That means that the information coming from this question is not very refined. Respondents who answer "good" are more positive than the people who said "not so good," but there is no information about the relative feelings of all the people who answer "good," even though there may be quite a bit of variation among them in the degree of positiveness they feel about the president's job.

There is another issue: the distribution of answers. In the above example, suppose most of the respondents answered the question in a particular way; for example, suppose 90% said the president was doing a "good" job. In that case, the value of the question is particularly minimal. The question gives us meaningful information for only about 10% of the population, the 10% who responded "not good." For the 90% of the population that answered "good," absolutely nothing was learned about where they stand compared with others who gave the same answer.

This analysis should suggest that there are two general principles for thinking about optimal categories for a response task. First, to the extent that valid information can be obtained, more categories are better than fewer categories. Second, generally speaking, an optimal set of categories along a continuum will maximize the extent to which people are distributed across the response categories.

Given these considerations, is there any limit to the number of categories that is useful? Is more categories always better? There are at least two limiting factors to the principle that using more categories produces better measurement. First, there appear to be real limits in the extent to which people can use scales to provide meaningful information. Although the optimal number of categories on a scale may vary, in part with the dimension and in part based on the distribution of people or items rated, most studies have shown that there is little new valid information provided by response tasks that provide more than 10 categories (Andrews, 1984). Beyond that, people seem not to be providing new information; the variation that is added seems to be mainly a reflection of the different ways the people use the scales. In fact, 5 to 7 categories is probably as many categories as most respondents can use meaningfully for most rating tasks.

A second issue has to do with the ease of administration. If survey instruments are being self-administered, where people read the questions to themselves, or are being administered in person, so that an interviewer can hand a respondent a list of the response categories, long lists of scale points do not pose any particular problem. However, when surveys are done on the telephone, it is necessary for respondents to retain all of the response options in order to answer the question. There clearly are limits to peoples' abilities to retain complex lists of categories.

When long, complex scales are presented by telephone, sometimes it is found that it produces biases simply because respondents cannot remember the categories well. For example, there is some tendency for respondents to remember the first or the last categories better than some of those in the middle (Schwartz & Hippler, 1991). When questions are to be used on the telephone, researchers often would prefer to use scales with only 3 or 4 response categories in order to ease the response task and ensure that respondents are aware of all the response alternatives when they answer questions.

A second independent issue is whether to use numerical or adjectival labels. The principal argument in favor of adjectival scales is that all of the points are more consistently calibrated by the use of words.

Consider the following question:

Example 3.11a: On a scale from 0 to 10, where 10 is as positive as you can be and 0 is as negative as you can be, how would you rate the movie you saw last night?

In such a response task, the two extremes probably are as well defined as they can be. However, there may be ambiguity about the meaning of "5." Is 5 a neutral point, where positive turns into negative? Does this mean that if someone's net feeling is more positive than negative, only scores of 6 or higher can be used? Alternatively, are people using the scale more like a thermometer or feeling ladder, really rating degrees of positiveness rather than going through a midpoint to a negative rating?

These are more than hypothetical questions.

Example 3.11b: Consider a scale from -5 to +5, where +5 is as positive as you can be and -5 is as negative as you can be. What number would you give to the movie you saw last night?

If 5 was consistently used as a neutral point in the first question, the one that went from 0 to 10, the results of these two questions should be identical. In fact, they are not identical. On average, people are more likely to report positive ratings using the second question than they are to report ratings of 6 or higher in response to the first question. Moreover, the distributions are not the same (Schwarz, Knauper, Hippler, Noelle-Neumann, & Clark, 1991). Almost certainly, people are inconsistent in the way that they use middle parts of the 10-point scale.

There is the potential to improve consistency by defining what meaning people should assign to the middle numbers. However, that sort of approach moves toward adjectival labeling of points. Probably, one can achieve more consistent use of scales by providing adjectives for all the response alternatives along a continuum.

The other side of the story is that it is difficult to think up adjectives for more than 5 or 6 points along most continua. When researchers have tried, some of the adjectival descriptions sound very close or similar to one another. It is virtually impossible to find a list of adjectives that will define a 10-point scale.

A related advantage of numbers is that a numerical 10-point scale is easy to remember and use. Thus, when doing telephone interviews, although it may be difficult to teach respondents 5 or 6 adjectives that they can reliably remember and use, it is comparatively easy to define a 10-point scale numerically. Hence, using scales defined by numbers can increase the reliability of a rating task performed on the telephone, if numerous response alternatives are to be provided. Moreover, it may increase the comparability of measurement of subjective ratings across modes of data collection.

Finally, a problem in international research and increasingly in research in the United States is how to get consistent measurement of subjective states for different cultural groups. In particular, when scales are defined adjectivally, it has been found that it is virtually impossible to have exact translations across languages. Adjectival scaling tasks across languages are not comparable.

It has not been well demonstrated that those problems are less severe if numerical scales are used. However, it certainly should be the case that such scales would translate better and more easily, because all that really is needed is a good translation of the two ends of the continuum.

In conclusion, asking people to answer rating questions using 2 or 3 categories constitutes an easier task for respondents and interviewers, but it provides less information per question. Using more categories, from 5 to 10, will provide more valid information about where people stand than rating tasks using fewer categories; more than 10 response categories probably adds very little valid information for most response tasks.

Finally, there are various occasions when using adjectival scales probably results in more consistent and therefore valid measurement. However, numerical response scales, with only the ends of the continuum defined, perhaps with some general discussion of how to use the points between the extremes, have numerous advantages and constitute a very good way to have people perform some rating tasks.

Using an Agree-Disagree Format

In the preceding sections, we have discussed asking respondents to classify or rate something by putting it on a continuum, often choosing an adjective or number to place it on the scale. The same kind of tasks can be attacked using an agree-disagree format:

The goal of such questions is basically the same as those discussed in the preceding section: to order people along some continuum.

Example 3.12a: I like Ike.

Example 3.12b: My health is excellent.

Consider the continuum from positive to negative that we have seen before. Both of these statements can be reliably located at the positive end of a continuum, either feelings about former president Eisenhower or the rating of a person's health.

Always	Usually	Often	Sometimes	Seldom	Never
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Figure 3.2. Frequency Scale

Suppose a person is asked to agree or disagree with these statements. In essence, respondents are saying that their views lie at the generally positive end of the continuum within a reasonable distance of where they see the statements "I like Ike" or "My health is excellent" to be located.

Example 3.13a: I am sometimes depressed.

Such a statement would be located in the middle of the continuum indicated in Figure 3.2. It is somewhere between the statement that "I am often depressed" and the statement that "I am seldom or never depressed." If a respondent is asked to agree or disagree with such a statement, there is an interpretation problem when a person disagrees. A person could disagree because "I am sometimes depressed" seems to understate the problem or because it seems to overstate the problem. As measurement, the results are unacceptable. To be able to interpret the answers, a different point on the scale must be chosen for the stem question:

Example 3.13b: I am usually depressed.

Example 3.13c: I am seldom depressed.

Using the agree-disagree or true-false strategy to put rated objects on scales has several drawbacks.

1. The items have to be unambiguously at an end of a continuum, so "disagree" answers can be interpreted unambiguously.
2. The questions are often cognitively complex; disagreeing that one is seldom depressed is a complicated way of saying one is often depressed.
3. Even when four categories are offered (such as strongly agree, agree, disagree, and strongly disagree), the answers usually are analyzed in two response categories: agree versus disagree. Hence, such questions do not yield much information.

4. There is a tendency among less-educated respondents toward acquiescence, which leads them to be particularly likely to answer in the "agree" direction. That tendency leads Converse and Presser (1986), among others, to urge avoidance of this question form.

For measuring respondents' feelings about ideas or policies, questions in this form are difficult to avoid. Such questions are discussed later in the chapter. However, if the goal is to have something rated or placed on a continuum, a more direct rating task will almost always accomplish the same thing better. We can ask people to agree or disagree with the statement, "my health is excellent." However, how much simpler, direct, and informative it is to ask: "How would you rate your health—excellent, very good, good, fair, or poor?"

Rank Ordering

There are occasions when researchers want respondents to compare objects on some dimension.

Example 3.14: Which candidate do you prefer?

Example 3.15: What do you consider to be the most important problem facing the city?

Example 3.16: Here are some factors some people consider when deciding where to live. Which is most important to you?

Proximity to work
Quality of schools
Parks
Safety
Access to shopping

Example 3.17: I'm going to read you a list of candidates. I want you to tell me which you consider to be the most liberal.

The basic question objectives can all be met in one of four ways:

Task 1. Respondents can be given a list of options and asked to rank order them from top to bottom on some continuum.

Task 2. Respondents can be given a list of options and asked to name the most (2nd most, 3rd most, etc.) extreme on the rating dimension.

Task 3. Respondents can be asked a series of paired comparisons, ordering two options at a time

Task 4. Respondents can be given a list and asked to rate each one using some scale (rather than just putting them in order or picking one or more of the most extreme).

If there is a short list of options, Task 1 is not hard to do. However, as the list becomes longer, the task is harder, soon becoming impossible on the telephone when respondents cannot see all the options.

Task 2 is easier than Task 1 when the list is long (or even when the list is short). Often researchers are satisfied to know which are the one or two most important, rather than having a complete rank ordering. In that case, Task 2 is attractive.

Psychometricians often like the paired comparison approach, by which each alternative is compared with every other, one pair at a time. However, it is such a time-consuming and cumbersome way to create an ordered list that it seldom is used in general surveys.

Best of all may be Task 4. The task is probably easiest of all for respondents, regardless of data collection mode. Moreover, the rank ordering tasks (Tasks 1 through 3) do not provide any information about where the items are located on the rating continuum. They could all be perceived as very high or very low, and the rank order provides no information.

Task 4 provides information about where the items are located on the rating scale. Although there can be ties, so ordering is not known perfectly, usually an aggregate order will result as well. For all these reasons, very often a series of ratings, rather than a rank order task, is the best way to achieve these objectives.

The Role of Open-Ended or Narrative Questions

When the goal is to place answers on a continuum, allowing people to answer in their own words will not do.

Consider a question such as:

Example 3.18: How is it going?

People can answer in all kinds of ways. Some will say "fine," some will say "great," some will say "not bad." If one were trying to order such comments, some ordinal properties would be clear. Those who say

"terrible" would obviously be placed at a different point on a continuum from those who say "great." However, there is no way to order responses such as "not bad," "pretty good," "good enough," or "satisfactory."

A first step for questions in this form is to define the continuum along which respondents are supposed to answer the question and to structure the response task, the kinds of answers that are acceptable, so that everyone is working with the same scale and the same response alternatives.

The continuum must be specified, and respondents must be told how to use it, by choosing among adjectives or numbers. When respondents are asked how they feel about something and left to use their own words, invariably they will use words that cannot be put in an ordered format. If ordering responses is the goal, narrative responses cannot be used.

In contrast, when the purpose of a question is to identify priorities or preferences among various items, there is a genuine choice.

Example 3.19: What do you consider to be the most important problem facing your city government today?

Example 3.19a: The following is a list of some of the problems that are facing your local city government:

- a. Crime
- b. Tax rates
- c. Schools
- d. Trash collection

Which do you consider to be most important?

The open-ended approach has several advantages. It does not limit answers to those the researcher thought of, so there is opportunity to learn the unexpected. It also requires no visual aids, so it works on the telephone. On the other hand, the diversity of answers may make the results hard to analyze. The more focused the question and the clearer the kind of answer desired, the more analyzable the answers. Moreover, Schuman and Presser (1981) found that the answers are probably more reliable and valid when a list is provided than when the question is asked in open form. If the list of possible answers is not known or is very long, the open form may be the right approach. Moreover, although computer-assisted interviewing creates great pressure to use only fixed-response

questions, respondents like to answer some questions in their own words. Although the measurement result may not be as easy to work with, asking some questions to be answered in narrative form may be justified for that reason alone. However, if good measurement is the goal and the alternatives can be specified, providing respondents with a list and having them choose is usually best. Appendix C presents further discussion of the use of open-ended questions.

Magnitude Estimation

The tasks discussed so far involved people using ordered categories or scales to rate their own subjective states or their perceptions of others. When defining response alternatives on scales, it is useful to use adjectives that can be differentiated and do not mean the same thing; however, for the measurement discussed so far, the distance between categories has not been assumed to have meaning. It has only been assumed that people in one category or scale point on a continuum are likely, on average, to differ from those who rated themselves either higher or lower.

Researchers would like to have people use ratings that provide more information than that. Even though it may be true that it is impossible to assign absolute meaning to a person's subjective state, researchers have explored the possibility of having people tell them something about the distance between categories or ratings beyond their ordinal position.

The technique that people have used that seems most successful at giving more absolute meanings to ratings is magnitude estimation. The following is an adaptation of how this approach was used to have people rate the social standing of people in different occupations (Rainwater, 1974).

Example 3.20: We are going to ask you to use numbers to describe the social standing of people doing different jobs. We are going to ask you to use the social standing of someone doing the work of a carpenter as a basis of comparison. Let us define the social standing of a person doing the work of a carpenter as 100. If you think the social standing of someone doing a different kind of job is twice as great as that of a carpenter, you would assign them a number of 200. If you thought the social standing of a person was half that of a person doing the work of a carpenter, you would assign them a number of 50. Using that approach, what number would you give to the social standing of a person doing the work of a high school teacher?

The purpose of this kind of exercise is to have people give numbers that have more absolute meaning and have some algebraic properties that most answers to questions about subjective states do not have. If we ask people to rate social standing on a scale from 1 to 10, those rated 6 would not be three times as good or high as those rated 2. People do not use that kind of scale in that way. However, the hope is that when given a magnitude estimation task, the ratings that people give do in fact have that kind of property. An interesting recent application of that strategy was to have physicians rate the difficulty and amount of work involved in various things that they do. The reason that the researchers wanted physicians to attempt to rate the difficulty of their work using magnitude estimation, rather than using a simple rating scale of the same thing, is that they wanted to relate the answers to another variable that does have absolute meaning: the amount of money physicians are paid for doing various services. The basic form of the questions asked was as follows: "If the work involved in performing an appendectomy is 100, what number would you assign the work involved in doing a cholecystectomy (gall bladder removal)?" In fact, the researchers studying the relative value of medical procedures and services mount compelling evidence that physicians were able to perform this task in a way that was consistent with providing real information about the relative value and difficulty of procedures compared with a standard (Hsiao et al., 1992).

Such an approach is not useful for measuring many of the subjective states that researchers want to measure. It takes a fair amount of training of respondents, which is time-consuming. Also, although the technique has been tried with cross-section samples, as was the case with Rainwater (1974), there clearly are advantages to having respondents, such as physicians, who are selected for their cognitive abilities (Schaeffer & Bradburn, 1989). As a result of all of these factors, magnitude estimation is not commonly used in surveys. Nonetheless, magnitude estimation is an example of how the measurement of subjective states can be stretched to provide more elaborate, and more analytically useful, information.

MEASURING RESPONSES TO IDEAS

The tasks discussed in the preceding section all were geared to having respondents place something on a rating scale or order items on a scale.

A large part of the survey research enterprise is focused on measuring people's responses to various ideas, analyses, or proposals. The content of such questions is as vast as the imagination of the survey research community. The common form of such questions is something like the following:

Example 3.21: Do you favor or oppose the idea of sending U.S. bombers to attack Antarctica?

Example 3.22: Higher taxes generally hurt the rich and benefit the poor. Do you agree or disagree?

Example 3.23: In general, would you like to have more money spent on the parks and playgrounds in your neighborhood area or not?

An important distinction to be made in thinking about questions like these is the nature of the task confronting the respondent. In the first section, respondents were asked to place themselves or others on some defined continuum. For example, they would be asked to rate their own health on a scale from excellent to poor or they would be asked to rate the job they thought that the president of the United States was doing from good to poor. The task posed by these questions, however, is somewhat different. Instead of placing some object on a defined continuum, the respondents are asked to rate the distance between their own views or preferences and the idea expressed in the question.

Consider the question about respondents favoring or opposing a bombing mission. Presumably, the respondent has some general opinions about how to approach the problem at hand. A policy alternative is outlined in the question. The respondents' job is to figure out whether or not the policy alternative is close enough to their own views that they can say that it is a "favored" policy.

In a parallel example, consider the proposition about taxes and the way they affect the rich and the poor. Obviously, this is a generalization. There is great potential for people to have perceptions that are more moderate or constitute only degrees of the extremity of the proposition as stated. Nonetheless, the task they are asked to perform is to look at their own views, in all of their simplicity or complexity, compare them with the statement, and decide whether the distance between the statement and their own views is close enough for them to say that they "agree."

The key difference to note is that respondents are not directly placing an object on a rating scale; rather, they are evaluating the distance

between their views and a statement. The standards for these questions are the same, though: Questions should be clear to all respondents and the response task should be one they are able to do.

Example 3.24a: Do you favor or oppose gun control laws?

Gun control laws can mean many things. They can cover a range of policies, including rules about who can buy guns, how long people have to wait to buy guns, and what kind of guns they can buy. A fundamental problem with questions such as this is that what respondents consider to be the meaning of "gun control laws" can differ from respondent to respondent.

To the extent that researchers can minimize differences in interpretation of what questions mean, they can increase the validity of measurement. The goal is to have differences in answers reflect differences in where people stand on the issues, rather than differences in their interpretations of the questions.

Example 3.24b: Do you favor or oppose laws that would prevent people convicted of violent crimes from purchasing a handgun or pistol?

Obviously that is only one kind of gun control law. However, a question that is explicit about the policy that people are being asked about, that minimizes differences in interpretation about what the question means, should produce more valid measurement of where people stand.

There are occasions when it is reasonable to let people define terms for themselves.

Example 3.25: Do you think help with personal or family problems should be covered by basic health insurance plans?

The term "personal or family problem" is not defined in this question. The rationale for asking this question this way is that it would be very difficult to define what constitutes a problem for a person. The nature of personal problems varies, and a situation that constitutes a problem for one person may not bother another person at all. In such cases, it may be best to let each respondent define what constitutes a problem. Nonetheless, from the point of view of interpreting the results, the researcher knows that there may be great heterogeneity in the objective situations that constitute the problems respondents are thinking about.

Consider another example:

Example 3.26: Driving a car is not a right, it's a privilege.

That may seem to be a reasonable statement to some. However, it is an example of a very common characteristic of questions posed in an agree-disagree format: It is two questions buried in a single question.

One question is: To what extent do you think that driving a car is a right?

The second question is: To what extent do you think driving a car is a privilege?

It may be that the answers to those two questions are negatively correlated; the less people think driving is a right, perhaps the more they think it is a privilege. However, putting the two issues together may make it hard to interpret what the answers mean. People might disagree because they think it is both a right and a privilege. Because there are two continua underlying this question, researchers cannot reliably place "disagrees" on either one of these continua.

It is very common to find multiple dimensions underlying questions posed in the agree-disagree format, or variations thereon. The following are examples, cited by Robinson and Shaver (1973), that have this characteristic:

Example 3.27: America is getting so far away from the true American way of life that force may be necessary to restore it.

Three issues: How far America is from the true American way, whether or not the true American way should be restored, and whether or not force may be needed (or desirable) to restore it.

Example 3.28: There is little use writing public officials because they often aren't really interested in the problems of the average man.

Two issues: Value of writing officials and how interested officials are in the problems of the average man.

Example 3.29: I feel completely secure in facing unknown new situations because my partner will never let me down.

Three issues: How secure one feels in new situations, the reliability of the partner, and whether or not the two are related.

Because there are at least two dimensions underlying these questions, when people answer the questions, the answers cannot be used reliably to order people.

Hence, there are two key generalizations about the kinds of questions that can be asked in this format. First, the words in the questions should be as well defined as possible to increase the consistency of respondent understanding of the ideas. Second, items should be carefully studied to make sure that only a single idea or question is being presented.

The Response Task

Probably the most common way to pose the agree-disagree task is the following:

Strongly agree
Agree
Disagree
Strongly disagree

Such a response task clearly violates the first rule of designing a response task for subjective questions: It contains two dimensions. The term "strongly" suggests an emotional component, a degree of conviction or caring about the answer over and above the cognitive task that is the central question. Better survey design would keep the response continuum cognitive.

There are two less frequently used approaches to the response task that avoid the emotional or affective component of the above scale and, therefore, are probably to be preferred.

Completely agree	Completely true
Generally agree	Mostly true
Generally disagree	Mostly untrue
Completely disagree	Completely untrue

Another response issue is whether or not to offer a middle category between agreeing and disagreeing and what it should include. There are two reasons why a person might not be able to agree or disagree. The position that would legitimately put someone in the middle of the

continuum described above is the person who perceived his or her opinions to be exactly in balance. Somehow, the person agreed to some degree, disagreed to the same degree, and could not reach a conclusion about the tilt. When respondents are forced to choose, by not being given the middle option, most of them do so. However, offering a middle option often is popular with respondents and conceptually it makes reasonable measurement sense.

There is another group of respondents that sometimes chooses this residual category: those who do not know enough about the question or their own opinions to answer the questions.

Example 3.30: The United States should increase its foreign aid to Guatemala.

It is likely that some respondents asked that question would not have enough information about either the U.S. aid program to Guatemala or the aid needs of Guatemala to have an informed opinion about that topic. In such a situation, for someone to say that they have no opinion is not to say that their opinions are balanced between agreeing and disagreeing; such respondents are off the scale. When respondents are likely not to know enough about a topic to answer, they should not be put in the middle category of the scale. Some systematic strategy should be found to identify people who lack the information needed to answer the question. That topic will be discussed in somewhat more detail in a subsequent section.

Measurement Properties of These Questions

Questions in the agree-disagree format are relatively indiscriminating questions. Most analysts divide respondents into two categories, those who agree and those who disagree. The reason for doing this is that there is research that shows that response style may have more to do with people's willingness to choose the extreme response than with differences in the opinions being reported. The issue of the limited amount of information that comes out of a two-category response task applies clearly to questions like these.

Cognitive Complexity

It is easy to write questions in this form that are very difficult to answer. Particularly when researchers try to ask questions about the negative ends of a continuum, it often produces cognitive complexity.

Example 3.31: I am not satisfied with my work.

In the agree-disagree format, in order to say that they are satisfied with their work, respondents have to disagree with such negative statements. Disagreeing with a statement that says they are "not satisfied" is a complex way to say one is satisfied. In cognitive testing, it is very common to find that respondents are confused by the task of how to communicate what they have to say in this format.

Conclusion

The agree-disagree question form, and its variants, is one of the most used measurement strategies in survey research. When asking about ideas or policies, it is appropriate. However, although such questions seem easy to write, they require considerable care to produce good measures.

There are three main problems with these questions. First, many questions in this form do not produce interpretable answers, either because they are not on a clearly defined place on a continuum or they reflect more than one dimension. Those problems can be solved by careful question design. However, two other problems—that these questions usually sort people into only two groups and that they often are cognitively complex—are more generic to the question form. Although it is possible to design good questions in this form, it often is possible to design a more direct rating task that will accomplish the same thing better.

MEASURING KNOWLEDGE

In survey research, knowledge is measured in four ways:

1. Asking people to self-report what they know
2. True-false questions
3. Multiple choice questions
4. Open-ended short-answer questions

Example 3.32a: Are you at all familiar with the proposals to control gun ownership?

Example 3.32b: How much do you know about the new proposal to control gun ownership—a lot, some, a little, or nothing at all?

Obviously, the goal of the questions is the key to the choice of questions. Probably the most common goal of knowledge questions in general population surveys is to identify those who think they are familiar enough with a topic to answer questions about it. A related common goal is to analyze whether those who feel informed about a topic think or behave differently from those who feel less informed. For those purposes, self-described knowledge is a reasonable approach to measurement. In fact, some researchers accomplish similar goals with questions that do not even refer directly to knowledge:

Example 3.32c: Have you heard or read about proposals to control gun ownership?

Example 3.32d: Have you discussed gun control with anyone?

Both of these questions can serve as indicators of familiarity with an issue. However, none of the questions noted so far has the potential to evaluate the quality or accuracy of information that people have. Sometimes researchers want to measure what people know.

Two favorite approaches to measuring knowledge are to use multiple choice or true-false questions. True-false and multiple choice questions share three features as measures of knowledge:

1. They measure recognition, rather than recall;
2. They depend heavily on constructing plausible wrong answers; and
3. The number of right answers usually is an overestimate of the number of questions to which respondents know the answers.

Recognition is easier than recall from a cognitive perspective. We all have had the experience of feeling we would know a name or a word if we heard it, but somehow cannot conjure it up without help.

The importance of plausible wrong answers is critical to understanding what the measurement process is about.

Example 3.33a: Who was the 13th president of the United States: Millard Fillmore, Zachary Taylor, or John Tyler?

Example 3.33b: Who was the 13th president of the United States: Millard Fillmore, Thomas Jefferson, or Richard Nixon?

The question and right answers are the same, but most people would find the second question easier than the first because of the wrong

options. Many more people would probably be able to rule out the wrong options to the second question than to the first.

Of course, neither question is as demanding as the open-ended form of the same question:

Example 3.33c: Who was the 13th president of the United States?

Example 3.33d: Which president was Millard Fillmore?

The strength of the open questions is that there are virtually no false positives; respondents either give the right answer or they do not. In contrast, in the multiple choice or true-false format, a random guess will pick the right answer at some rate. Thus, if 60% of the people answer a true-false question correctly, one might conclude that only 20% of the people actually gave the right answer because they knew it; the other 40% (equal to the wrong answers) could have given the right answer by chance.

The disadvantage of the open form of the question is that it may provide a low estimate of active knowledge, because some people who could recognize the correct answer, or retrieve it given more time, will fail to retrieve it in a survey situation. Moreover, the level of specificity required affects the results. There are many people who could correctly identify Fillmore as a 19th century president and who would get no credit in response to 3.33c or 3.33d.

Finally, three general points that apply to all survey questions apply particularly to measures of knowledge. First, for the short-answer question form, it is particularly important that what constitutes an adequate answer is clearly specified by the question. To the extent that deciphering what kind of answer is wanted affects the correctness of answers, the question will be a less valid measure of knowledge. Second, measures of knowledge are question-specific, just like measures of other subjective states. The same level of knowledge may produce different results depending on the question. Third, the value of a measure of knowledge is usually dependent on how well answers are distributed. If most people are in the same category (right or wrong), a question doesn't give the researcher much information.

MULTI-ITEM MEASURES

One of the important ways to improve the measurement of subjective states is to combine the answers from more than one question into an

index (DeVellis, 1991; Nunnally, 1978). There are at least two reasons why multi-item indices can produce better measurement than a single item:

1. They can produce detailed measurement across a larger spectrum of a continuum than a single question (or do it with less burden on respondents).
2. By diluting item-specific effects, they can produce a better measure of what a set of items has in common.

Improving Discrimination

One of the earliest forms of multi-item measures was the so-called Guttman Scale. The following is an example:

1. Can you walk across a small room?
2. Can you walk up stairs?
3. Can you walk a city block?
4. Can you walk half a mile?
5. Can you run half a mile without stopping?

If the above questions form a perfect Guttman Scale, without exception, the number of "yes" answers would tell a researcher which of the things on the list a respondent could or could not do. If these items scale, everyone who can run half a mile can do all of the preceding things; if a person can climb a flight of stairs but not walk a city block, the person should also be able to walk across a small room, if the scale is perfect.

Obviously, the basic concept is that there is a dimension of a physical ability. Ideally, each of these questions defines a point along that continuum. By asking a set of "yes or no" questions, then looking at the answers as a group, one can figure out where along the continuum the person falls.

A slightly different application of the same principle can be used that does not assume the kind of order among items that the Guttman approach assumes. Suppose a health condition can affect people in various ways that may not necessarily be related. For example, arthritis could make it difficult to:

1. walk up the stairs
2. use a pencil
3. bend or stoop
4. reach high shelves.

There is no necessary order among these various ways that arthritis might affect function; problems with using a pencil are likely to be unrelated to problems with bending or stooping. However, it might make sense to ask questions about the variety of ways that function might be affected by arthritis, then combine the answers in some way to create a symptom severity index.

There are issues about how to combine items like this, particularly when they are not necessarily correlated and are reflecting different aspects or manifestations of the underlying condition. Should these all be weighted equally, or are some symptoms more important than others? DeVellis (1991) presents a much more elaborate discussion of how to combine items and the value thereof; however, the basic point is that asking a series of questions, then combining them, sometimes is the best way to produce a measure of a complex state, such as a health condition.

Measuring a Common Underlying Variable

Any time one is using a question to measure some dimension, the answer is likely to reflect the dimension the researcher is trying to measure plus some other things. Suppose we were trying to create a measure of the way that a health condition affects people's lives. The following questions are asked:

1. To what extent does your condition limit the amount or kinds of work you can do at your job?
2. To what extent does your condition limit the amount or kinds of things you do for fun and recreation?
3. To what extent does your condition limit the amount and kind of work you do around the house?
4. To what extent does your condition limit your ability to get around and to go where you want to go?

The purpose of these questions is to get a measure of the severity of the effect of some health condition. However, the answer to each of these questions is likely to reflect both the severity of the symptoms from the condition and the particular style of the individual or the demands on the individual in the various domains asked about. For example, the effect of a health condition on a person's ability to work might depend importantly on the kind of work a person does (for example, how much physical effort is involved) as well as on the severity of the condition. In a parallel way, the effect on recreation

might depend, in part, on the kinds of recreational activities a person prefers. The avid card player or television watcher might be less limited than the avid swimmer or hiker by the same condition.

The point is that the answer to each of these questions is likely to be affected both by what we are trying to measure (the severity of the condition) and by the particular role expectations or lifestyle of the respondent. However, the unrelated components of the answers are also likely not to be related across questions. Hence, just because a person has a job that requires physical effort does not mean that he or she has active tastes in recreation or in the kinds of things enjoyed around the house. By combining the answers to these questions, it is likely that an index can be built that is less affected by roles than any particular question and hence more purely a measure of condition severity.

To take another example, suppose our goal is to measure psychological distress. The most common forms of psychological distress of which people are aware are depression and anxiety. However, people have different words that they use to code or describe these states. For depression, people use "sad," "blue," and possibly "down in the dumps." For anxiety, people use "anxious," "worried," "nervous," "tense," "flustered," and so forth. For any particular person, with some set of experiences, some of these words may capture the way they experience their psychological distress better than others. By asking several different questions, using various of these words, it is likely that researchers can produce an index that captures psychological distress better than any single question, with whatever idiosyncrasies of interpretation or experience the particular words chosen bring with them.

When items are combined to produce a multi-item index, the index measures whatever it is that the items have in common. A multi-item scale is no guarantee that an index is measuring well what a researcher wants to measure. That has to be demonstrated in other ways. It does ensure, however, that whatever the items have in common is measured better than it would be with a single item alone (Cronbach, 1951).

THE RELATIVITY OF ANSWERS ABOUT SUBJECTIVE STATES

The answers to questions about subjective states are always relative; they are never absolute. The kinds of statements that are justified, based on answers to these kinds of questions, are comparative. It is appropriate to say that Group A reports more positive feelings than Group B. It

is appropriate to say that the population reports more positive feelings now than it did a year ago. It is not appropriate (at least not without some careful caveats) to say that people gave the president a positive rating, that they are satisfied with their schools, or that by and large they think their health is good.

In Chapter 2, the discussion of questions about objective facts noted that there are standards for right and wrong answers. Although sometimes the questions and definitions were complex, what was meant by being the victim of a burglary could be defined, and the accuracy of reports could be checked against records. If events are defined clearly, definitions are communicated in a consistent way, and questions are asked that people are able and willing to answer, the answers have absolute meaning.

In contrast, when people use a rating scale to evaluate their fear of crime or their feelings about a political candidate, there is not the same kind of absolute meaning about what a "good" answer means. Furthermore, the proof is that the answers that result from surveys are critically affected by many of the details of the way the question is worded and asked. The properties of the stimulus question have a critical effect on the response. The meaning of a response can be interpreted only in the context of knowing what the stimulus was.

Question Wording

If the answers to subjective questions were to have absolute meaning, the answers to cognitively equivalent questions should be the same. Often that is true. For example, it has been shown that substituting "surgically end a pregnancy" for the term "abortion" does not change people's answers to opinion questions on that topic (Schuman & Presser, 1981). Apparently, those words mean the same thing to people. However, Rasinski (1989) showed that people were much more enthusiastic about increased spending on people with "low incomes" than increased spending on people on "welfare." Schuman and Presser (1981) found that nearly half the population would support "not allowing" Communists to speak in public, but only about 20 would "forbid" it. The "meaning" of questions obviously can go beyond the literal interpretation of the words.

There are numerous examples of how small changes in wording, which seem to be equivalent, produce very different results. That is one of the reasons that interviewers are taught to read questions exactly as worded. Questions that seem to be the same often are not the same from the point of view of being a stimulus to generate a response about a subjective state.

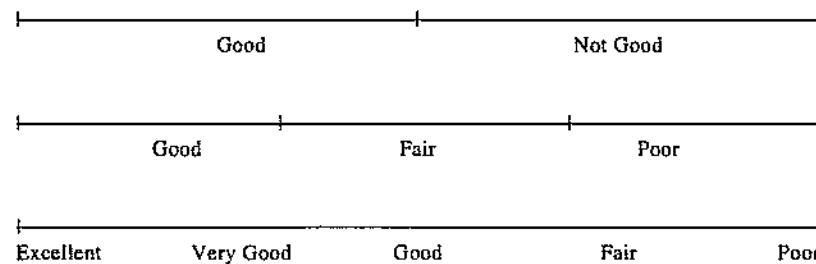


Figure 3.3. Three Scales

The Wording of Response Alternatives

The fact that the wording of response alternatives changes the distribution of responses is also clear evidence of the relativity of the measurement of subjective states.

Consider Figure 3.3, which defines an evaluative rating scale. On the top line, the scale is divided into two categories, "good" and "not good." On the second line, the scale is divided into three categories: good, fair, and poor. On the third line, the scale is broken into five categories: excellent, very good, good, fair, poor.

Someone might think that Scale 2 would simply divide the right-hand category, "not good," from the first scale into two parts. Similarly, one might think that Scale 3 would divide "good" from Scales 1 and 2 into three new parts. However, as some readers might already know, that is not how it works. People consider the whole scale when thinking about the meaning of categories. They consider the ordinal position of the category, as well as the meaning of the words. As a result, there will be many more people who rate their health as "good" or better in response to Scale 3 than do so in response to Scale 2. For that reason, stating that 70% of the population rates their health as "good" or better is an uninterpretable statement out of the context of the particular question that people were asked.

Order of Alternatives

The order in which response alternatives are presented to people also can have some bearing on how they answer questions. This is particularly true when respondents are given lists of response preferences that

are not on an ordered scale. For example, consider a survey prior to an election primary. Respondents are read a list of several candidates, then asked which one they are most likely to vote for at the moment. Admittedly, an important characteristic of this particular survey task is that respondents may not be thoroughly familiar with all primary candidates. Nonetheless, it has been shown that when such surveys are done, the candidate whose name is read last disproportionately benefits and is disproportionately selected.

Even more subtly, it has been shown that the order of the basic rating scale, from excellent to poor, will affect the distribution of answers. When the alternatives are listed from poor to excellent rather than the other way around, respondents are more likely to use the negative end of the scale (Dillman & Tarnai, 1991).

Mode of Data Collection

For many purposes, it has been found that people answer questions in comparable ways by telephone, to a personal interviewer, or in a self-administered form (Fowler, 1993). However, there is some evidence that the way the question is presented, the mode of data collection, affects the way people use scales. For example, Bishop, Hippler, Schwartz, and Strack (1988) found that respondents answering self-administered questions used extreme response categories more often than those who were responding to an interviewer.

Context Effects

The location of a question in a survey instrument can also affect the way in which it is answered. Numerous studies have shown that the content of preceding questions can affect the way respondents interpret and answer individual questions (Groves, 1989; Schuman & Presser, 1981; T. W. Smith, 1991; Turner & Martin, 1984).

How "Don't Knows" Are Handled

A final important issue about the relativity of survey responses is the way that researchers handle the possibility that respondents are not familiar enough with the subject matter, or their own feelings on a topic, to answer a question. There are three different ways that researchers handle the possibility of lack of respondent familiarity with a topic. First, they may ask a screening question, explicitly asking respondents

if they are familiar enough with a particular topic to answer a question about it. Second, researchers can include a "no opinion" option among the response alternatives provided to respondents. Third, the researchers can choose not to explicitly address the issue of respondents' familiarity with a topic; they can force respondents who do not feel they can answer a question to volunteer that information.

It is not surprising to find out that the results are quite different, depending on which of these options researchers choose. There is reluctance on the part of respondents to volunteer that they do not know enough about a topic to answer a question. As a result, many more respondents will choose that option if it is made explicit than will volunteer it. Moreover, in some cases, the distribution of answers obtained is different, depending on whether respondents are or are not explicitly offered the option of saying they do not have an opinion on a topic (Schuman & Presser, 1981).

Conclusion

It is important to differentiate between effects of question form on the distribution of answers and effects on the validity of the answers. For example, it was noted that when asking people to use the 5-point rating scale from excellent to poor, respondents were more likely to use the categories at the negative end of the scale if the negative end of the scale was presented first. This would change the mean or average rating to a lower level, but it might not change the validity or value of the question as a measure. The ratings emerging from the two question forms might correlate equally well with other measures with which they should correlate, which in turn means they are equally valid.

Another implication of this point is that the concept of "bias" in a measure of a subjective state essentially is meaningless. By changing wording, response order, or other things about data collection, it is possible to change the distribution of answers to a question in a positive or negative direction. However, the concept of bias implies systematic deviations from some true score. There is no true score in the measurement of subjective states, so the fact that a distribution is changed in one direction or another does not constitute bias. Moreover, if these ordinal measures are used properly, to compare distributions or parameters among groups from whom data were collected in a comparable way, the concept of bias in measurement is indeed irrelevant.

One of the most common abuses of survey measurement is people treating data collected using measures of subjective states, that are

designed to produce ordinal measures, as if they had produced data with absolute meaning. When statements are made, such as that "most people favor gun control," "most people oppose abortion," or "most people support the president," the statements should be viewed askance. All that happened in any of these cases was that a majority of respondents picked response alternatives to a particular question that the researcher chose to interpret as favorable or positive. That same group of people could be presented with different stimuli that apparently addressed the same topic but would produce very different distributions and support very different statements.

CONCLUSION

In conclusion, there are four basic messages about how to design questions to measure subjective states.

1. Because there are no standards against which to evaluate the correctness or rightness of answers, standardization of the stimulus of the question is particularly critical in measuring subjective states. For this reason, designing questions that can be administered in a consistent way and that mean the same thing to all respondents, to the extent possible, is high on the list of strategies for creating good measurement of subjective states.
2. Equally important is standardizing the response task. That means clearly defining the dimension or continuum respondents are to use in their rating task and giving them a reasonable way to place themselves, or whatever else they are rating, on that continuum.
3. The distribution of answers that come from a question about a subjective state is relative; it has no absolute meaning. Hence, the mean or average answer to a question is irrelevant, indeed meaningless. Rather, the standard for questions about subjective states is the extent to which they provide valid information about the position of respondents in relation to other respondents. Generally speaking, the more categories respondents are asked to use, the better.
4. Combining the answers to several questions often is an effective way to increase the validity of measurement.

Having enunciated these various principles, in the next chapter we will discuss some specific strategies for designing survey instruments that provide good, useful measurement of subjective states and objective facts.